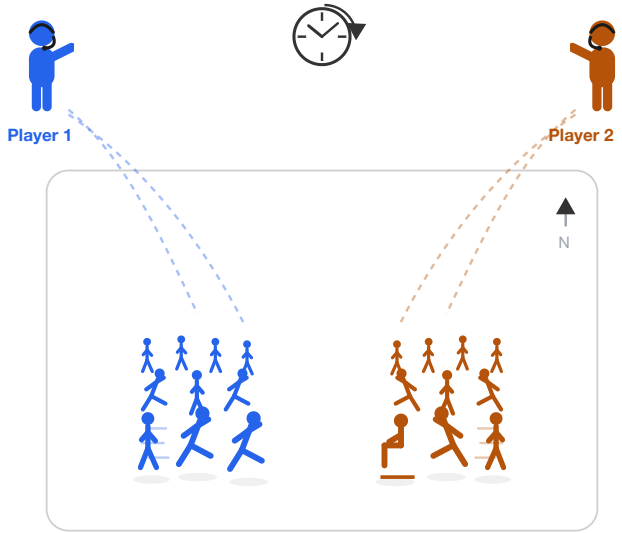
World Commander

Natural-language command of agent crowds in strategy games, under real-time compute budgets



What we build: real-time, language-commanded agent crowds (here, two players commanding their own).

The North Star: the next revolutionary video game, played by voice from the commander's chair.

This proposal grew from Yubo Huang's interest and insight, under the guidance of Dr. Enmao Diao.

June 2026

1.1 The Problem and the Vision

- In a strategy game, the **fun is the strategy**: where to expand, when to attack, how to read the opponent.
- It is buried under **frantic clicks**: hunting icons, drag-selecting, hundreds of actions a minute.
- A player should command like a **real commander**: watch, listen, **speak the orders**, press a key only now and then.
- The subordinates execute. Strategy is where we start; the interface could be the next revolutionary game, in any genre.

Figure 1: the labour the interface forces. A few StarCraft II tasks (move here, gather there, build these); the strategy is buried under hundreds of manual actions a minute.

Still: PySC2 video (DeepMind)

1.2 The Gap: Efficiency, Untested at Game Speed

- The interface shipped once (Tom Clancy's EndWar (2008)), but on a 70-word grammar. Language models lifted that limit; they still **cannot act at game pace**.
- LLMs already play RTS through text (TextStarCraft II (NeurIPS 2024)), but only by **pausing the clock**.
- The efficient-inference methods that would fix that (eviction, quantization, distillation, smaller models) are tuned on perplexity, **never in a live game** where dropping the wrong context loses the match.
- A streaming game is the natural stress test, and **nobody has run it** (LLM Game Agents Survey (CSUR 2026) names it a core open challenge).

> **How much does command cost, in latency and memory, at game speed, and how do we drive it down?**

2 Preliminaries

The background this work builds on:

Area	What it gives us
Reinforcement learning	A game is a Markov decision process: an agent acts, and win rate is the reward. We need to *understand* this to read the field; the work runs models by prompting, training none of its own.
Efficient LLM inference	KV-cache eviction (H2O, SnapKV, StreamingLLM, OBCache), structured pruning, quantization, distillation: the methods this work puts to the test.
Vision-language-action models	π0: a slow vision-language backbone driving a fast action expert, trained by imitation. A model for splitting a slow strategic brain from fast executors.
The StarCraft II agent stack	PySC2, TextStarCraft II, LLM-PySC2: the environment we inherit rather than rebuild.
Tokenization beyond text	Byte-pair encoding and graph tokenization (ICLR 2026): the basis for a learned game-state tokenizer.

3.1 The Three Jobs

The human commands; the system needs three capabilities to carry that out. Separate lines of work that eventually run together.

Job	What it does
Execute	An LLM turns the commander's spoken intent into the right in-game actions, in real time, under hard latency and memory budgets
Foresee	A fast world model the commander can query before committing: if the army pushes now, does the fight win?
Embody	Units carry out the actions with model-generated motion, at crowd scale, on one GPU

This proposal is about the first job, **Execute**. Foresee and Embody come later.

3.2 The Plan: Three Phases

The jobs are the *parts*; the phases are the *order*. One shared **harness** runs under all of them (command protocol, unpausable clock, deadlines, metric logging), so the engineering transfers upward.

Phase	Focus	What happens
1	Benchmarks	The command arena (warm-up), then StarCraft II with the clock unpaused.
2	Methods	Attack whatever the benchmarks expose: game-aware eviction, a learned tokenizer, distilled commanders.
3	Real interface	The human (voice), then humans plural (multiplayer); grow the Foresee and Embody jobs.

Environments grow with the phases: **a toy room → StarCraft II → multiplayer → a full game.** The end-state game is the north star, not a deliverable.

4.1 The Command Arena

- Colour-tagged agents in a room; each moves in **one of four directions** on command.
- One command is trivial; the test is the **stream**: many, fast.
- Harder with **rate**, **compositional orders** ("everyone but the yellow one"), and **memory** ("the one I moved west").
- Measured: **grounding accuracy**, **command-to-action latency**, **deadline misses**.

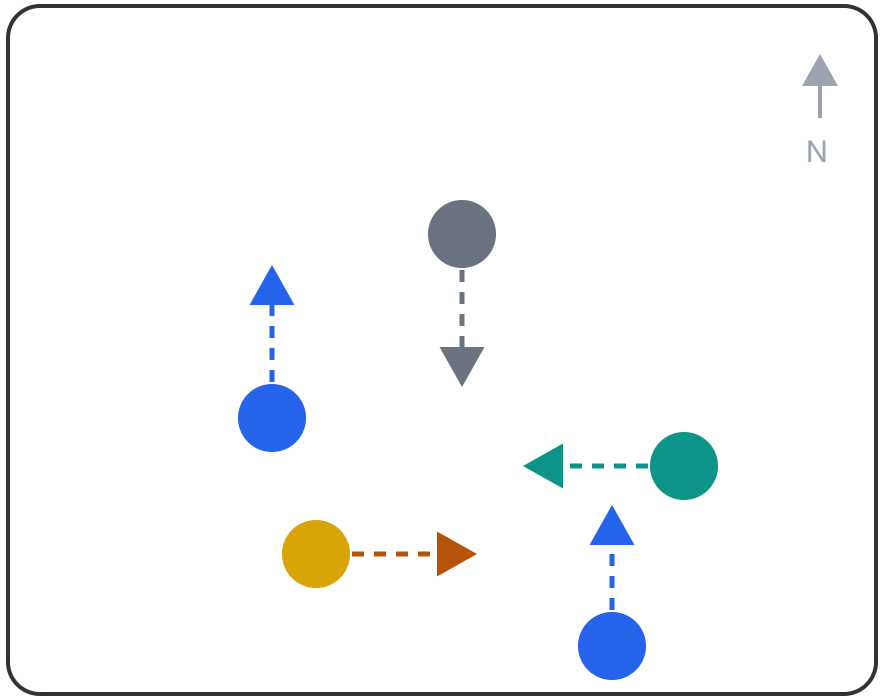


Figure 2: the command arena. Colour-tagged agents, each moving in one of four directions on command.

4.2 Then StarCraft II

- Once the arena works, the same harness moves to **StarCraft II with the clock unpaused**: real strategic depth, a real opponent.
- The LLM executes a **fixed, scripted strategy**; the efficiency methods (cache policy, model size, state encoding) are scored by **win rate under a latency and memory budget**.
- Concrete benchmark design waits until the arena is built.

5 What This Produces

The game is the north star; the papers are the milestones. Each answers a question its community already cares about, beyond the game.

Paper	Phase	The question	Who cares
Command-arena benchmark	1	How does grounding degrade as command rate rises?	Real-time agents, interactive systems
Real-time commander benchmark	1	Which efficiency methods survive a closed-loop game clock?	KV-cache and pruning researchers
Game-state tokenizer	1 to 2	Does compact tokenization extend to entity and event streams?	Tokenization beyond text
Competitive-efficiency study	3	Does a fast small model beat a slow large one?	Inference efficiency, agents
Crowd motion under budget	3	Can language-commanded crowds move in real time on one GPU?	Motion generation, graphics